

AuraRad / X-Ray Disease Prediction - Complete Interview Preparation Guide

This guide contains the most important interview questions and easy-to-speak answers for your X-Ray Disease Prediction / AuraRad project. The answers are written in simple and natural English so you can confidently explain your project in technical, HR, and real-world interview rounds.

1. Project Introduction

Q: How should you introduce this project?

A: One of my major projects was an X-ray Disease Prediction system for pneumonia detection using deep learning. Initially, it started as a CNN-based image classification model, but later I enhanced it into an Explainable AI system. The model predicts whether a chest X-ray is Normal or Pneumonia, and I implemented Grad-CAM heatmaps to visually explain which regions influenced the prediction. I used TensorFlow and Keras for the deep learning model, FastAPI for the backend, and Next.js with Tailwind CSS for the frontend interface. I also integrated a local GPT4All assistant to provide simplified explanations while maintaining patient data privacy. This project helped me gain practical experience in medical imaging, Explainable AI, and end-to-end AI system integration.

Q: Why did you build this project?

A: I built this project because early pneumonia detection is very important in healthcare. I wanted to create a system that can assist doctors by analyzing chest X-rays automatically. I was also interested in learning how deep learning and Explainable AI can be applied in medical imaging.

2. Tools, Technologies, and Algorithms

Q: Which technologies did you use?

A: I used Python as the main programming language. TensorFlow and Keras were used for building the CNN model. OpenCV was used for image preprocessing. FastAPI was used for backend API development. Next.js and Tailwind CSS were used for the frontend. GPT4All was used to build the local chatbot assistant.

Q: Which algorithm did you use?

A: I used a Convolutional Neural Network (CNN) because CNNs are highly effective for image classification tasks like X-ray analysis.

Q: Why CNN?

A: CNN automatically learns image features such as edges, textures, and infected regions from X-ray images.

3. Dataset and Training

Q: Which dataset did you use?

A: I used the Chest X-ray Pneumonia dataset from Kaggle containing Normal and Pneumonia X-ray images.

Q: How did you preprocess the images?

A: I resized the images to a fixed size, normalized pixel values, and applied augmentation techniques like rotation, zooming, and flipping.

Q: How did you split the dataset?

A: I divided the dataset into training, validation, and testing sets approximately in a 70:15:15 ratio.

Q: How did you prevent overfitting?

A: I used dropout layers, data augmentation, validation monitoring, and early stopping to reduce overfitting.

4. Explainable AI and Grad-CAM

Q: What is Explainable AI?

A: Explainable AI helps humans understand how AI models make decisions instead of behaving like black boxes.

Q: Why is Explainable AI important in healthcare?

A: Doctors need transparency before trusting AI predictions. Explainability helps improve trust and validation.

Q: What is Grad-CAM?

A: Grad-CAM generates heatmaps showing which regions of the X-ray influenced the model's prediction.

Q: How does Grad-CAM work?

A: Grad-CAM calculates gradients of the predicted class with respect to the feature maps of the last convolutional layer. These gradients are combined to create a heatmap highlighting important regions in the image.

5. Evaluation Metrics

Q: Why is accuracy alone not enough?

A: Medical datasets are often imbalanced. A model can get high accuracy by predicting Normal most of the time, but that can still be dangerous.

Q: What is Recall?

A: Recall means out of all actual pneumonia patients, how many were correctly identified by the model. In healthcare, Recall is very important because missing a sick patient can be dangerous.

Q: What is Precision?

A: Precision means out of all the patients predicted as pneumonia, how many were actually correct.

Q: What is F1-score?

A: F1-score is the balance between Precision and Recall.

Q: What is a Confusion Matrix?

A: A Confusion Matrix shows True Positives, True Negatives, False Positives, and False Negatives. It helps understand where the model makes mistakes.

6. Backend and Frontend

Q: Why FastAPI instead of Flask?

A: FastAPI provides better performance, asynchronous support, automatic validation, and auto-generated API documentation.

Q: How does frontend communicate with backend?

A: The frontend communicates with the backend using REST APIs and HTTP requests.

Q: Why Next.js?

A: Next.js provides better routing, fast loading, and a modern frontend structure.

Q: Why Tailwind CSS?

A: Tailwind CSS helps build modern user interfaces quickly using utility-based styling.

7. Chatbot and Privacy

Q: How did you build the chatbot?

A: I integrated GPT4All as a local Large Language Model through the backend.

Q: Why use a local LLM?

A: Using a local LLM helps maintain patient privacy because medical data does not leave the system.

Q: How does the chatbot work?

A: The chatbot processes user questions locally and generates simple explanations related to the diagnosis.

8. Real-world and Behavioral Questions

Q: Can AI replace doctors?

A: No. AI should assist doctors, not replace them. Final diagnosis should always be made by medical professionals.

Q: What if the model predicts a sick patient as healthy?

A: That is called a False Negative, which is dangerous in healthcare. That is why Recall is very important in medical AI.

Q: What was the biggest challenge?

A: Integrating TensorFlow inference and the local LLM while keeping the system responsive was one of the biggest challenges.

Q: What did you learn from this project?

A: This project improved my understanding of deep learning, Explainable AI, backend integration, and real-world AI system design.

Q: What would you improve in the future?

A: I would improve the model using larger datasets, stronger transfer learning architectures, and better clinical validation.

9. Basic Concepts You Must Know

Q: What is Deep Learning?

A: Deep learning is a subset of machine learning that uses neural networks with multiple layers to learn patterns from data.

Q: What is CNN?

A: CNN is a deep learning model mainly used for image processing and classification.

Q: What is Convolution?

A: Convolution is the process where filters slide over the image to detect important features.

Q: What is Pooling?

A: Pooling reduces image size while preserving important features.

Q: What is ReLU?

A: ReLU is an activation function that helps the model learn non-linear patterns.

Q: What is Overfitting?

A: Overfitting happens when the model performs well on training data but poorly on new unseen data.

Q: What is Data Augmentation?

A: Data augmentation artificially increases dataset diversity using transformations like rotation and flipping.

Final Interview Tips:

- Speak slowly and confidently.
- Explain concepts in simple English.

- Focus on understanding, not memorization.
- In healthcare AI, always remember: “AI should assist doctors, not replace them.”